

MATHEMATICS

TARGET : JEE- Advanced

CAPS-1**FUNDAMENTAL OF MATHEMATICS &
TRIGONOMETRY****SCQ (Single Correct Type) :**

- The number of solutions of the system of equations

$$x + \sqrt{x^2 + 1} = 10^{y-x}$$

$$y + \sqrt{y^2 + 1} = 10^{z-y}$$

$$z + \sqrt{z^2 + 1} = 10^{x-z}$$
 are
 (A) 4 (B) 3 (C) 2 (D) 1
- Total number of values of x , of the form $\frac{1}{n}, n \in \mathbb{N}$ in the interval $x \in \left[\frac{1}{25}, \frac{1}{10}\right]$, which satisfy the equation $\{x\} + \{2x\} + \dots + \{12x\} = 78x$, (where $\{\cdot\}$ represents fractional part function)
 (A) 12 (B) 13 (C) 14 (D) 15
- The value of $\sum_{K=1}^7 \tan^2\left(\frac{K\pi}{16}\right)$ is ____
 (A) $\frac{3}{2}$ (B) 105 (C) 35 (D) 21
- If $\frac{x^2 - 2|x| - 3}{x^2 - 4|x| - 5} > -1$, then x equal to
 (A) $(-\infty, -5) \cup (5, \infty)$ (B) $(-4, 4)$
 (C) $(-\infty, -5) \cup (-4, 4) \cup (5, \infty)$ (D) $\mathbb{R}$
- In $\triangle ABC$, if $\sin 2A + \sin 2B + \sin 2C = 4 - 4 \cos A \cos B$, then triangle is
 (A) equilateral (B) right angled but not isosceles
 (C) isosceles but not right angled (D) isosceles and right angled

MCQ (One or more than one correct) :

- If $f(\theta) = \sum_{n=1}^6 \operatorname{cosec}\left(\theta + \frac{(n-1)\pi}{4}\right) \operatorname{cosec}\left(\theta + \frac{n\pi}{4}\right)$, where $0 < \theta < \frac{\pi}{2}$, then minimum value of f
 (A) lies between 3 and 4 (B) lies between 2 and 3
 (C) occurs when $\theta = \frac{\pi}{4}$ (D) occurs when $\theta = \frac{\pi}{6}$
- Let $E = \cos^2 \frac{\pi}{7} + \cos^2 \frac{2\pi}{7} + \cos^2 \frac{3\pi}{7}$. Then which of the following alternative(s) is/are incorrect?
 (A) $\frac{1}{2} < E < \frac{3}{4}$ (B) $\frac{3}{4} < E < 1$ (C) $1 < E < \frac{3}{2}$ (D) $\frac{3}{2} < E < \frac{7}{4}$

8. Consider the equation $\sin\left(\pi \log_3\left(\frac{1}{x}\right)\right) = 0$ Then which of the following statements is/are true ?
- (A) there exists, infinitely many solutions for the equation in $(0, 2\pi)$
 (B) there exists, finitely many solutions for the equation in $(0, 2\pi)$
 (C) $x = 3$ is one of the solution for the given equation in $(0, 2\pi)$
 (D) sum of all the solutions of the given equation in $(0, 2\pi)$ is $\frac{9}{2}$
9. If $S(n) = \sum_{r=1}^n \frac{\cos(2 \times 3^{r-1} \alpha)}{\sin(3^r \alpha)}$, $\alpha = 18^\circ$ and $P(n) = \sum_{r=1}^n \frac{\cos(3 \cdot 2^{r-1} \beta)}{\cos(2^r \beta)}$, $\beta = 36^\circ$, then
- (A) $S(2012) + P(2012) = 2010$ (B) $S(2012) + P(2012) = 2012$
 (C) $P(2011) - S(2011) = 2009$ (D) $P(2011) - S(2011) = 2008$
10. If $\sin x + \cos x + \tan x + \cot x + \sec x + \csc x = 7$ and $\sin 2x = a - b\sqrt{7}$, then $a - b + 14$ is divisible by ____.
- (A) 2 (B) 3 (C) 4 (D) 7
11. Let $\frac{b \cos x}{2 \cos 2x - 1} = \frac{b + \sin x}{(\cos^2 x - 3 \sin^2 x) \tan x}$, $b \in \mathbb{R}$. Which of the following are correct?
- (A) The given equation has a solution if $b \in \left(-\infty, \frac{1}{2}\right) - \left\{-1, 0, \frac{1}{3}\right\}$
 (B) The given equation has a solution if $b \in \left(\frac{1}{2}, \infty\right) - \left\{1, \frac{1}{3}\right\}$
 (C) The given equation has no solution if $b = 10$
 (D) The given equation has no solution if $b = -10$
12. If $(\sin 2x + \sqrt{3} \cos 2x)^2 - 5 = \cos\left(2x - \frac{\pi}{6}\right)$ then the value of $\cos x$ can be
- (A) $\frac{\sqrt{6} + \sqrt{2}}{4}$ (B) $\frac{\sqrt{6} - \sqrt{2}}{4}$ (C) $-\left(\frac{\sqrt{6} + \sqrt{2}}{4}\right)$ (D) $\frac{\sqrt{2} - \sqrt{6}}{4}$
13. Possible integral value(s) of m for the equation $\sin x - \sqrt{3} \cos x = \frac{4m - 6}{4 - m}$ can be valid for some $x \in [0, 2\pi]$ is
- (A) -1 (B) 0 (C) 1 (D) 2

Numerical based Questions :

14. If complete solution set of inequality $\sqrt{x^2 - 2x} > 4 - x$, is $\left(\frac{m}{n}, \infty\right)$, m, n being co-prime numbers, then value of $m + n$ is
15. The number of solutions of equation $8[x^2 - x] + 4[x] = 13 + 12[\sin x]$ is (here $[.]$ represents greatest integer less than or equal to ' x ')
 16. If $3p \sin x - (p + \sin x)(p^2 - p \sin x + \sin^2 x) = 1$ has a solution for x , then the number of integral values of p are _____

17. If α and β are roots of equation $(\log_2 x)^2 + 4(\log_2 x) - 1 = 0$, then value of $\log_\beta \alpha + \log_\alpha \beta$ equals
18. Let $x = (\log_{1/3} 5) (\log_{125} 343) (\log_{49} 729)$ and $y = 25^{3 \log_{289} 11 \log_{28} \sqrt{17} \log_{1331} 784}$, then find the value of $x^2 + y^2$ is
19. Let $f(x) = |x - 2|$ and $g(x) = |3 - x|$ and
 A be the number of real solutions of the equation $f(x) = g(x)$
 B be the minimum value of $h(x) = f(x) + g(x)$
 C be the area of triangle formed by $f(x) = |x - 2|$, $g(x) = |3 - x|$ and x-axis and $\alpha < \gamma < \beta < \delta$ where $\alpha < \beta$ are the roots of $f(x) = 4$ and $\gamma < \delta$ are the roots of $g(x) = 4$, then find the value of sum of digits of $\frac{\alpha^2 + \beta^2 + \gamma^2 + \delta^2}{ABC}$.
20. The reciprocal of the value of the product $\sin\left(\frac{5\pi}{2^{11}}\right) \cos\left(\frac{5\pi}{2^{11}}\right) \cos\left(\frac{5\pi}{2^{10}}\right) \cos\left(\frac{5\pi}{2^9}\right) \dots \cos\left(\frac{5\pi}{2^3}\right) \cos\left(\frac{5\pi}{2^2}\right)$ equals.

Comprehension Type Question:**Paragraph for question nos. 21 to 23**

Let A denotes the sum of the roots of the equation $\frac{1}{5 - 4 \log_4 x} + \frac{4}{1 + \log_4 x} = 3$.

B denotes the value of the product of m and n, if $2^m = 3$ and $3^n = 4$.

C denotes the sum of the integral roots of the equation $\log_3 x \left(\frac{3}{x}\right) + (\log_3 x)^2 = 1$.

21. The value of A + B equals
 (A) 10 (B) 6 (C) 8 (D) 4
22. The value of B + C equals
 (A) 6 (B) 2 (C) 4 (D) 8
23. The value of A + C - B equals
 (A) 5 (B) 8 (C) 7 (D) 4

Matrix Match Type :

24. Match the following :

Column-I		Column-II	
A	The number of real roots of the equation $3^{ x } 2 - x = 1$ is /are	p	3
B	The number of real roots of the equation $3 \cos^{-1} x - \pi x - \frac{\pi}{2} = 0$ is / are	q	4
C	The number of real roots of $\sin x \cos y = 1 \forall x, y \in [0, 2\pi]$ is / are	r	2
D	The least degree of polynomial with integer coefficient one of the roots may be $\cos 12^\circ$ is/are	s	1

Code :

- (A) $A \rightarrow (q)$; $B \rightarrow (p)$; $C \rightarrow (p)$; $D \rightarrow (q)$
 (B) $A \rightarrow (q)$; $B \rightarrow (s)$; $C \rightarrow (q)$; $D \rightarrow (q)$
 (C) $A \rightarrow (q)$; $B \rightarrow (s)$; $C \rightarrow (p)$; $D \rightarrow (q)$
 (D) $A \rightarrow (p)$; $B \rightarrow (s)$; $C \rightarrow (p)$; $D \rightarrow (q)$

25. Math the following :

Column-I		Column-II	
A	If $\alpha, \beta (\alpha < \beta)$ are the two roots of the equation $\frac{1 - 8(\log_{10} x)^2}{\log_{10} x - 2(\log_{10} x)^2} = 1$ then the value of $\frac{(\beta^3 + 1)}{\alpha^2 \beta^3}$	P	5
B	Number of solution(s) of equation $\log_2(x^2 + 3) = \frac{1}{2} \log_{1/3} \left(x + \frac{1}{x} \right), x > 0$ is	Q	11
C	Number of integers satisfying the equation $ x + \left \frac{4 - x^2}{x} \right = \left \frac{4}{x} \right $ is	R	0
D	If $\log_c 2 \log_b 125 = \log_{10} 8 \log_c 10$ where $c > 0, c \neq 1, b > 0, b \neq 1$, then b is	S	4

Code:

- (A) A - P; B - S; C - R; D - Q
 (B) A - Q; B - S; C - R; D - P
 (C) A - P; B - R; C - S; D - Q
 (D) A - Q; B - R; C - S; D - P

ANSWER KEY (CAPS-1)

1. (D) 2. (B) 3. (C) 4. (C) 5. (D) 6. (BC) 7. (ABD)
 8. (ACD) 9. (AD) 10. (ACD) 11. (AC) 12. (BD) 13. (ABCD) 14. 11.00
 15. 0.00 16. 4.00 17. 18.00 18. 34 19. 9 20. 1024 21. (C)
 22. (A) 23. (B) 24. (C) 25. (D)